Dashboard user guide

Start

From the project folder on Windows (no need to activate the environment or change the PowerShell execution policy):

```
.\.venv\Scripts\python.exe -m streamlit run app/Home.py --server.address 127.0.0.1 --server.headless
true --browser.gatherUsageStats false
```

Open http://127.0.0.1:8501. The dashboard reads the latest pipeline run; it never retrains. On Windows, double-clicking run_dashboard.bat in the project folder does the same. On first use it also creates the environment, which needs Python 3.13 and an internet connection.

Single-file edition (HTML)

The same dashboard also comes as one file: reports/dashboard/retention-dashboard.html, which is 04_Dashboard/dashboard_html/Retention_Dashboard.html in the submission folder.

- **Opening it.** Double-click the file to open it in Chrome, Edge or Firefox. It needs no Python, no server and no internet connection.
- **What it contains.** It embeds the results of the pipeline run named in its sidebar: the scores of the held-out test customers, the model evidence and the deployed model.
- **What it computes.** It computes every campaign, KPI, rule comparison, break-even, chance of a loss and scenario analysis in the browser, with the same rules as the pipeline. tests/test_html_dashboard.py checks that it produces the same lists and numbers.
- **How it differs from the Streamlit dashboard:**
- The chance of a loss uses the browser's own random draws, so it can differ by about a percentage point.
- There is no scenario ID.
- The export is a set of CSV and JSON files instead of a ZIP.
- **Scoring new customers.** An uploaded CSV is scored in the browser by the same model, and nothing is sent anywhere.

Layout

- **Sidebar:** page navigation, the customer list in use (held-out test customers, or an uploaded CSV once scored) and the deployed model.
- **Assumption strip** (top of the *Decision* and *Assumptions & sensitivity* pages): budget, standard offer and its cost, contact cost, save rate, margin, value horizon and discount rate. The run and **scenario ID** are under *Technical details* on the Decision page and in every export; with default assumptions the ID equals the one in the reports.
- Budget and offer policy can be changed on the Decision page; every other assumption on the *Assumptions & sensitivity* page. Changes last for the session and carry across pages; the configuration file is never modified.

Pages

1. **Decision** - answers the business question. Five KPI cards (expected net value with the difference vs risk-only targeting, ROI, customers contacted, precision among contacted, high-risk/high-value coverage), the recommended action with its break-even save rate, **who gets an offer and who does not** (profile of the list, risk x value segments compared with risk-only targeting, and the risk x value map), a same-budget comparison of six rules, the **action list** (search, contacted-only filter, per-customer explanation) and the **export** button.
2. **Assumptions & sensitivity** - every assumption, the expected net value and break-even save rate of the current list, and the **chance this list loses money** when all register assumptions are drawn at once (live, 2,000 draws). Press *Run scenario analysis* to compute the tornado chart, pessimistic/base/optimistic scenarios, profit curve with marginal ROI and the save rate x offer cost grid (frozen or re-optimised list); results are cached per scenario and a warning asks to re-run after changes. The page ends with the pipeline's stress test (lower churn, Vietnamese prices).
3. **Model evidence** - four tabs: *Performance* (test metrics of the deployed calibrated logistic model with bootstrap intervals, calibration and gains, all candidates), *Drivers and scorecard* (odds ratios, the points card with score bands, and the XGBoost challenger with SHAP, clearly marked as not deployed), *Errors and money check* (each candidate valued in money, who the model gets wrong, performance by segment) and *Data and new customers* (source, quality and join, why churners left, **Score new customers**). This page computes no campaign.

Offer policy

- **Single standard offer:** every customer can only receive the standard credit.

- **Segment-specific offers:** each customer may receive one offer whose eligibility rule they meet (contract upgrade for month-to-month, service bundle for internet customers without TechSupport, loyalty reward for tenure > 24 months, standard credit for everyone). The optimiser chooses the mix that maximises total expected value within the budget and the contact cap. This is the default when it won on validation.

Uploading customers

- UTF-8 (with or without BOM), comma-separated, up to 10 MB / 50,000 rows; use the template and the data dictionary.
- customerID must be unique; Churn, gender and SeniorCitizen are optional and never used for scoring or selection.
- Missing numbers are filled with training medians. Invalid numbers, unknown categories, missing columns, duplicate or empty IDs, or service contradictions (for example PhoneService = No with MultipleLines = Yes) reject the whole file with a row/column error table; the previous customer list stays in place.
- Origin shows whether an uploaded ID matches the project's train, validation or test customers; results on IDs seen in training are not an independent evaluation.
- After scoring, choose *Uploaded CSV* in the sidebar. Without labels, precision among contacted is shown as the expected value (mean predicted probability).

Reading the results

- KPIs add up contacted customers only; other rows show the best potential offer for reference.
- The per-customer explanation shows p, save rate, value, cost, EV and break-even save rate - an economic explanation, not a cause of churn.
- In the grid, the save-rate axis is the standard offer's rate; other offers move by the same percentage points. The cost multiplier applies to offer costs, not contact cost.
- A frozen list can exceed the budget or lose money when costs rise or save rates fall; that is a robustness check, not a feasible new list.
- The export ZIP contains the full targeting table, the contact list, assumptions.json (with scenario ID) and summary.json. It contains no contact details and sends nothing.

Troubleshooting

Symptom	Fix
No module named telco_retention	Use the .venv interpreter and run pip install --no-deps -e .
"Analysis results are not ready"	Run python -m telco_retention run-all
Port 8501 in use	Close the previous session or add --server.port 8502
CSV rejected	Export UTF-8 comma-separated, follow the template, read the error table
Raw checksum mismatch	The raw file changed; restore it - the pipeline never overwrites raw data
Scenario charts out of date	Press *Run scenario analysis* again after changing assumptions
Chance of a loss shows 0% with nobody selected	No customer has positive expected value under these assumptions: not running the campaign is the right decision
PDF export fails	Microsoft Word/PowerPoint are needed for PDF; run export --pdf skip for DOCX/PPTX only